

BASIC PRINCIPLES OF MEDICINE 1: **FOUNDATIONS TO MEDICINE**

LECTURE FTM 9

Post Transcriptional Modification

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Objectives: DLA: RNA Polymerase and Transcription

- Before lecture 'Post-transcriptional modifications'

SOM.MK.I.BPM1.1.FTM.3.BCHM.0049	Describe the structural features of a gene and their functions
SOM.MK.I.BPM1.1.FTM.3.BCHM.0050	Describe the types and functions of the different cellular RNAs (mRNA, tRNA, rRNA, snRNA, miRNA)
SOM.MK.I.BPM1.1.FTM.3.BCHM.0051	Describe the major steps of RNA synthesis in prokaryotes and eukaryotes
SOM.MK.I.BPM1.1.FTM.3.BCHM.0052	Explain the function and structure of DNA sequences that are involved in the initiation (promoters) and termination of transcription
SOM.MK.I.BPM1.1.FTM.3.BCHM.0053	Predict the sequence of the transcribed RNA if the template DNA strand or the coding strand of the RNA is provided
SOM.MK.I.BPM1.1.FTM.3.BCHM.0054	List the RNA products that are transcribed by the three different types of RNA polymerases in eukaryotes
SOM.MK.I.BPM1.1.FTM.3.BCHM.0057	Explain the mechanism of action of rifampin and alpha-amanitin as inhibitors of transcription.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0058	Describe the effect of the antibiotic drugs that interfere with DNA gyrase such as (quinolones) ciprofloxacin

Objectives: Post transcriptional modification

SOM.MK.I.BPM1.1.FTM.3.BCHM.0059	Differentiate between the structures of prokaryotic and eukaryotic mRNAs
SOM.MK.I.BPM1.1.FTM.3.BCHM.0060	Describe the post-transcriptional modifications of eukaryote mRNA and the significance of these modifications (5' capping, polyadenylation, and splicing)
SOM.MK.I.BPM1.1.FTM.3.BCHM.0061	Predict effects of splice site mutations on mature mRNA and ultimately the translated protein
SOM.MK.I.BPM1.1.FTM.3.BCHM.0062	Describe how RNA editing affects our understanding of transmission of genetic information from the genome to protein
SOM.MK.I.BPM1.1.FTM.3.BCHM.0063	Describe human disease conditions that may be associated with mRNA modification and errors in this process

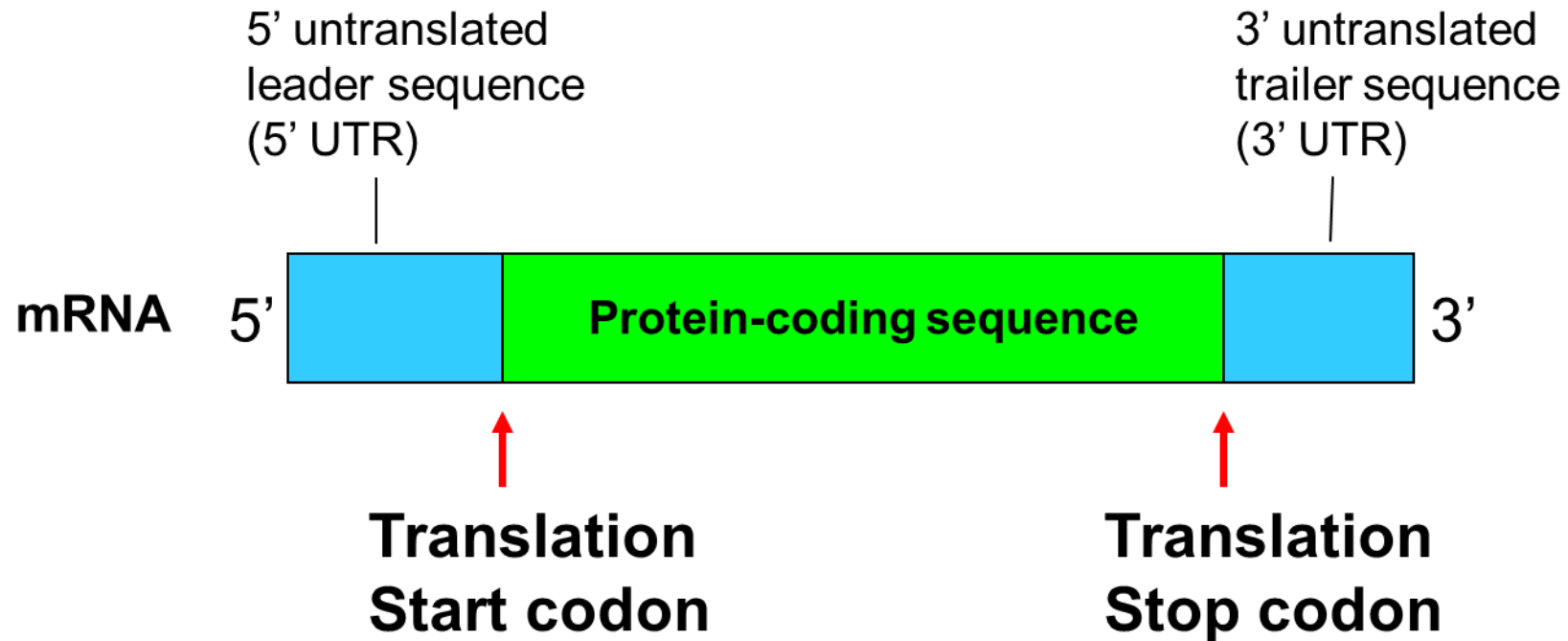
Pre-lecture Reading

View Panopto videos:
Transcription
Inhibitors of Transcription

Recommended Reading

- **Transcription and Post-transcriptional Modification Study Notes**
- Lippincott's Biochemistry 9th Edition: Chapter 31: RNA Structure, Synthesis and Processing: <https://premiumbasicsciences.lwwhealthlibrary.com/content.aspx?sectionid=260214583&bookid=3402>
- Lippincott's Biochemistry Chapter 31: [RNA Structure, Synthesis, and Processing | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library](#)
- Questions: 31.1 – 31.5

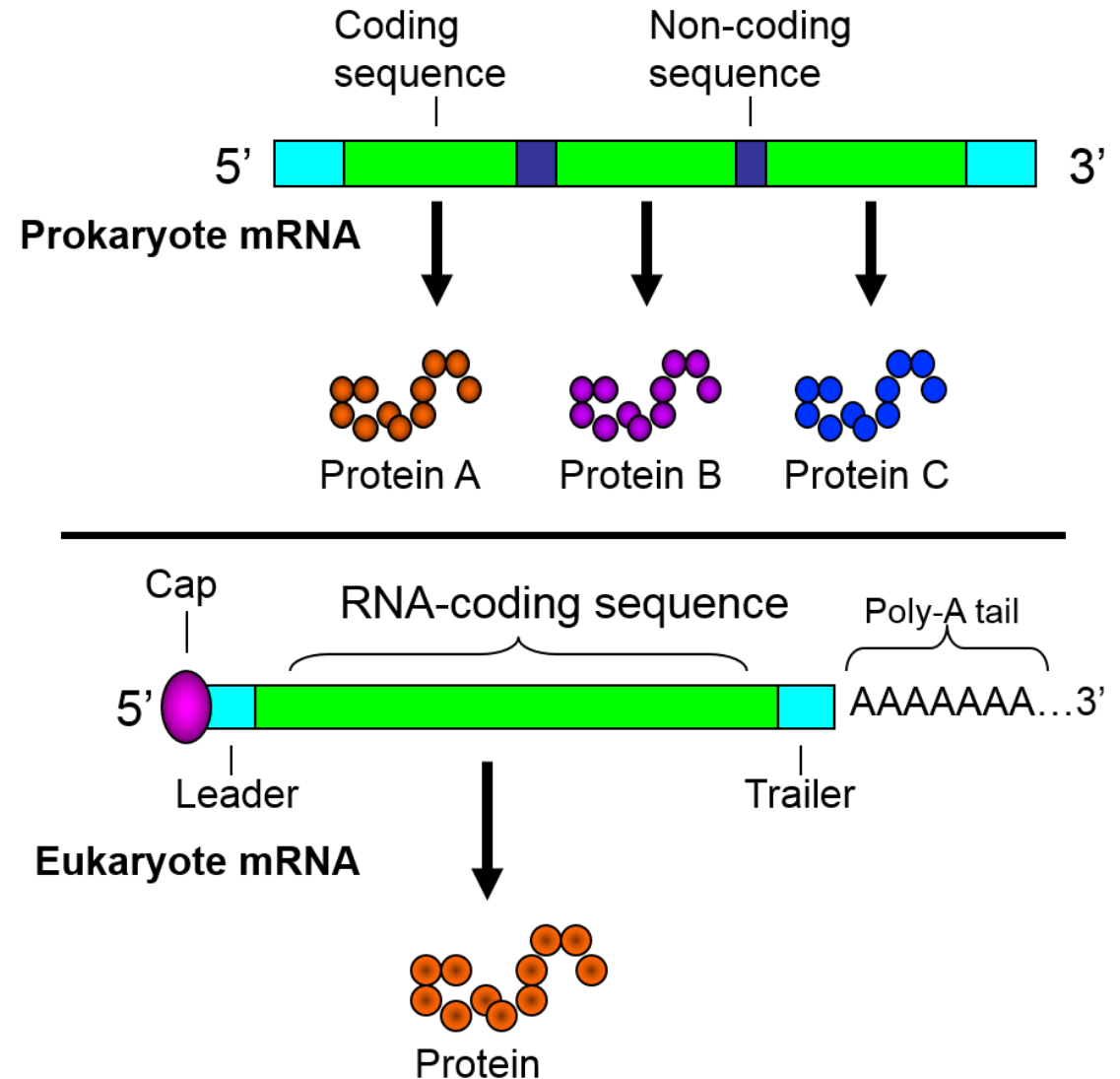
General structure of mRNA found in both prokaryotic and eukaryotic cells



- **UTRs** occur in the mRNA from the transcription initiation to the translation start site (5' UTR) and the translation stop site and the transcription terminator (3' UTR) during transcription

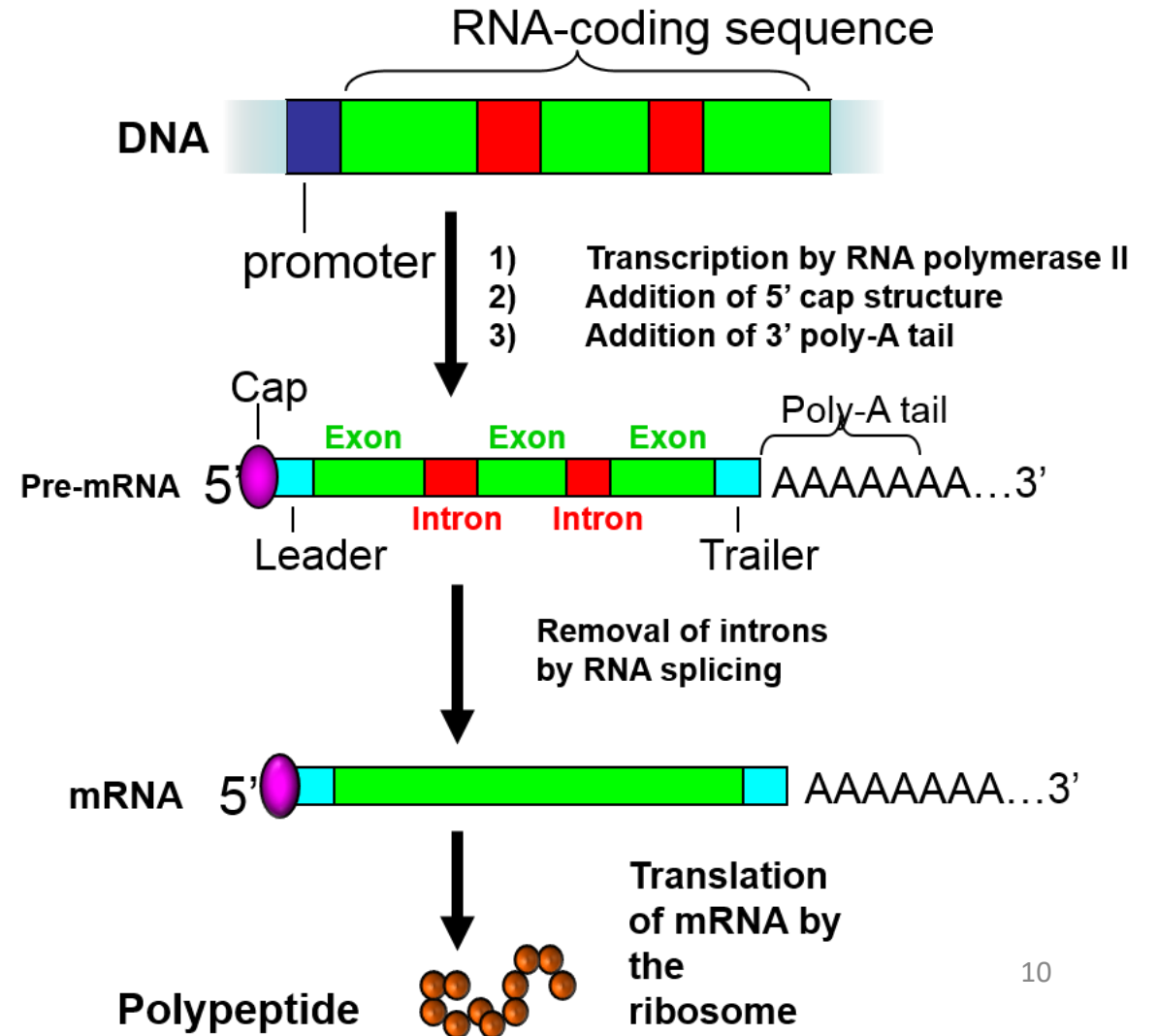
Comparison of mRNA structure: Prokaryotes vs. Eukaryotes

- The 5' and 3' end of bacterial mRNA are unmodified
 - Prokaryotic mRNA may be polycistronic
- Eukaryotic mRNA contains modification:
 - 7-methylguanosine cap on the 5' end
 - Poly-adenylation at the 3' end



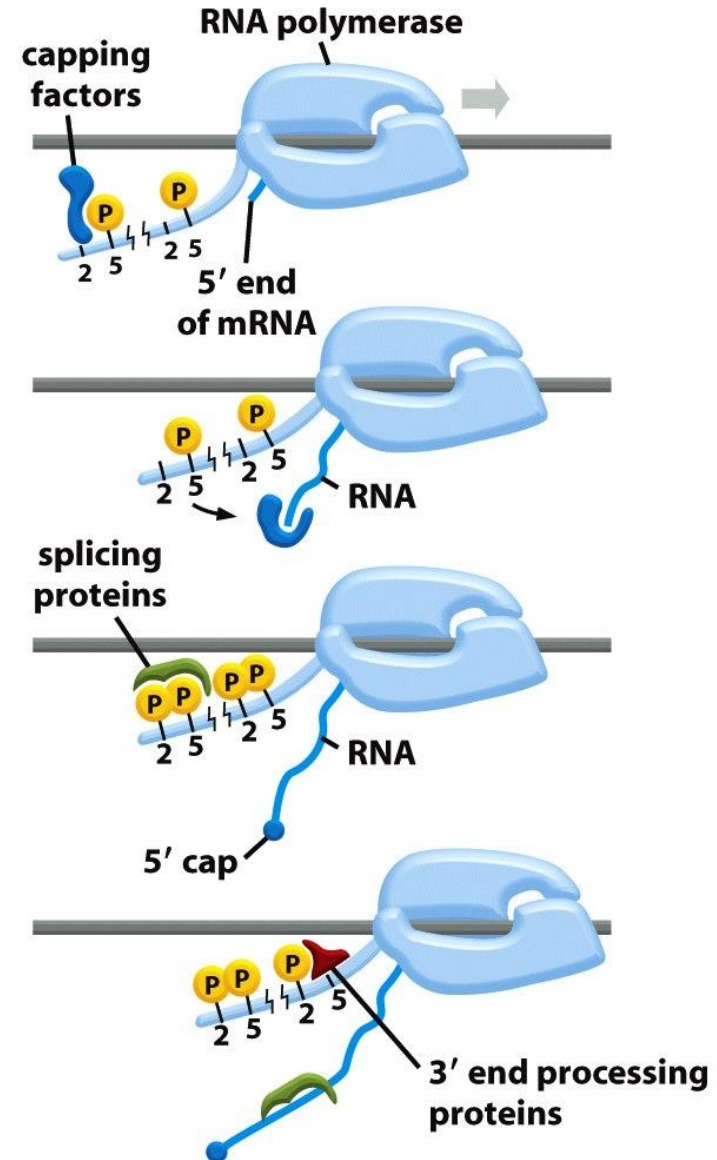
In Eukaryotes, the initial product of transcription must be processed and modified to form the mature mRNA

- Modification of the 5' end of the mRNA
- Polyadenylation of the 3' end of the mRNA
- Splicing of exons and the removal of introns



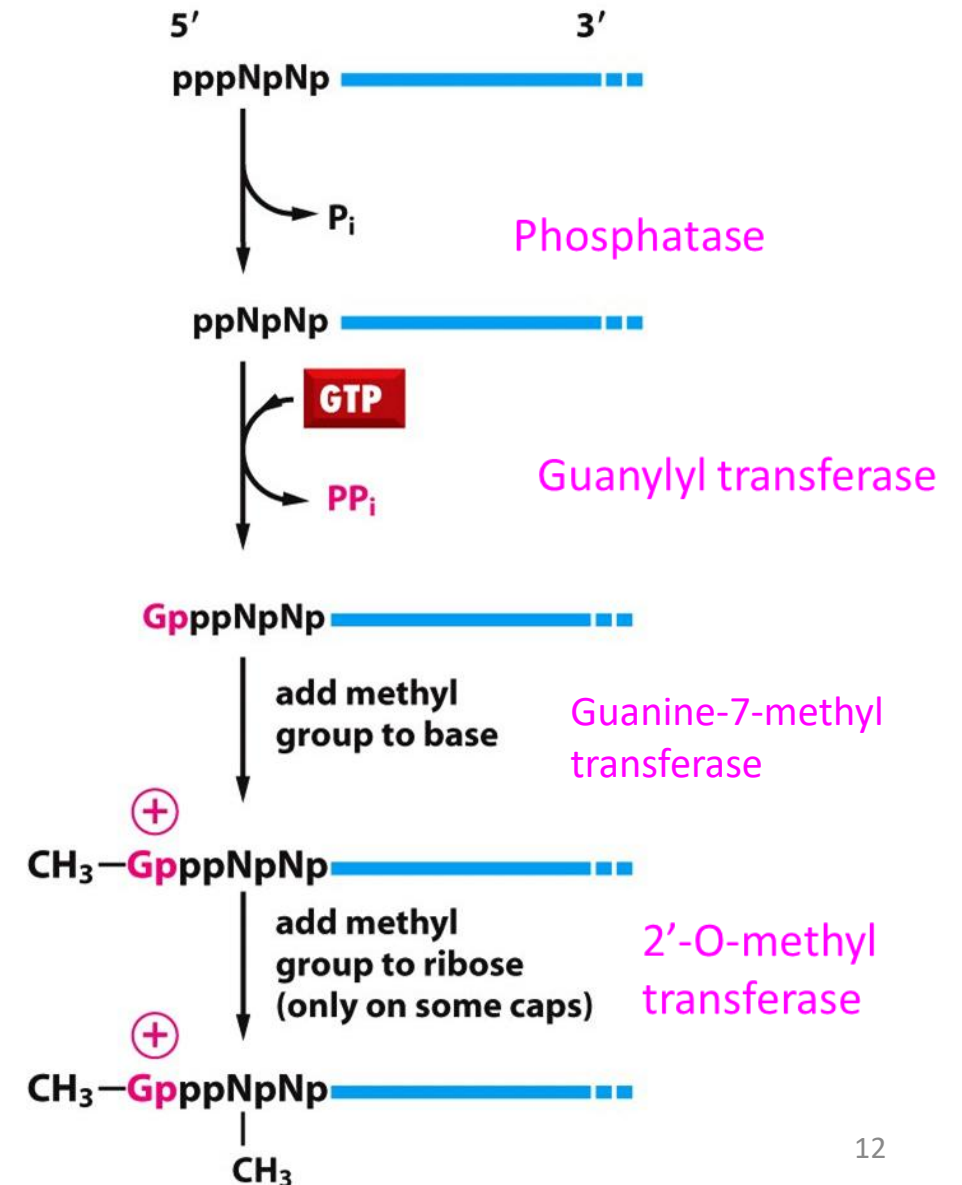
RNA polymerase II carries a set of pre-mRNA processing proteins on its C-terminal tail

- Shortly after the initiation of transcription, proteins are transferred to the nascent RNA (newly formed RNA)
 - Capping factors
 - Splicing factors
 - Polyadenylation factors
- This transfer of RNA processing proteins begins after approximately 25 nucleotides of the mRNA are produced



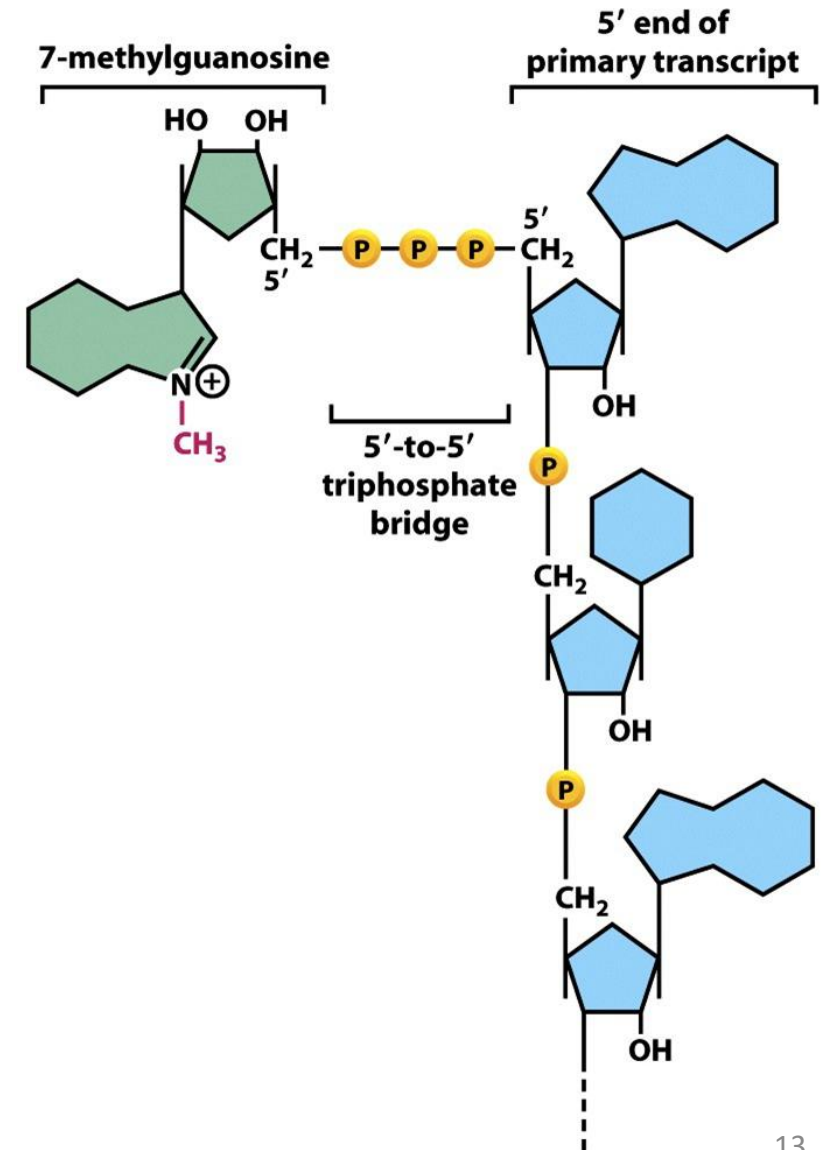
Capping of the 5' end of nascent mRNA: performed by four enzymes.

- 1) **Phosphatase:** removes one phosphate from the 5' end of the RNA
- 2) **Guanylyl transferase:** adds a GMP in a reverse linkage (5' to 5' instead of 5' to 3')
- 3) **Guanine-7-methyl transferase:** adds a methyl group to the 7 position of the terminal guanine
- 4) **2'-O-methyl transferase:** adds a methyl group to the 2'-O position to the next to last base on the 5' end



Close-up of the 5' cap on mRNA transcripts produced by RNA-polymerase II

- 5'-methyl cap is specific for eukaryotic mRNA and helps cell to distinguish between different RNA in the cell
 - mRNA vs. tRNA vs. rRNA
- 5'-methyl cap has important roles in the regulation of mRNA:
 - Processing
 - Transport
 - Translation



Functions of 5' mRNA capping

- Regulates export of mRNA out of the nucleus.

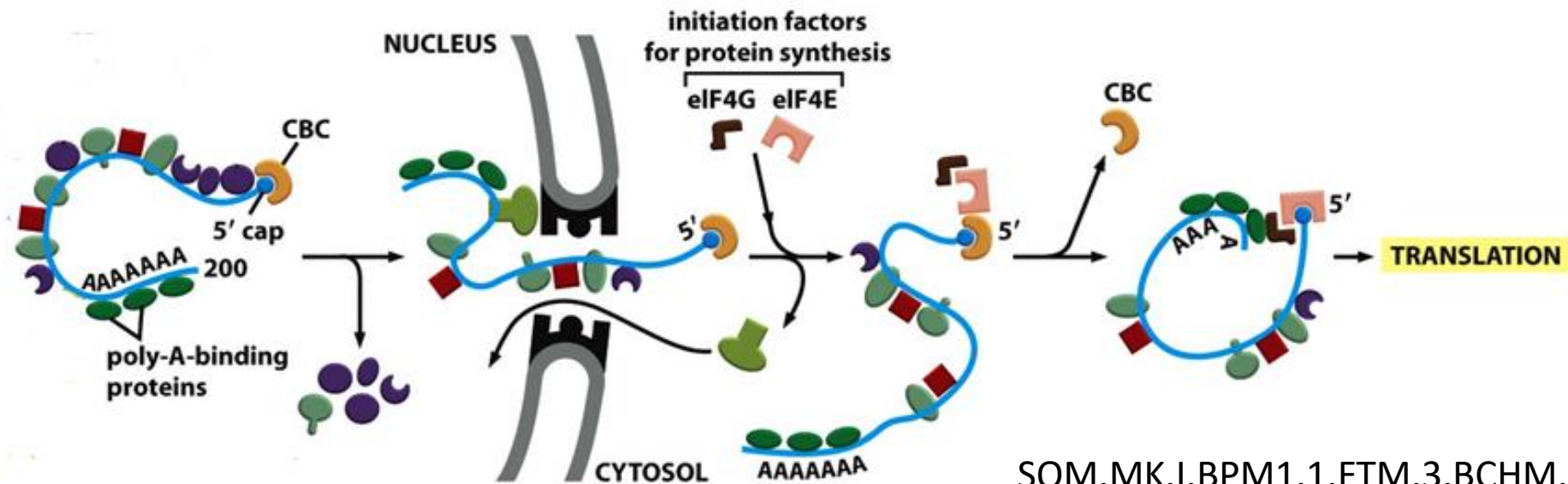
mRNA is exported in complexes that contain a Cap Binding Complex (CBC) at the 5' end & RNA binding proteins along the rest of the sequence.

- Required for the efficient translation of the mRNA into protein.

The CBC is replaced by the translation factors eIF-4E & eIF-4G which facilitates binding of mRNA to the ribosome.

- Prevents 5' degradation.

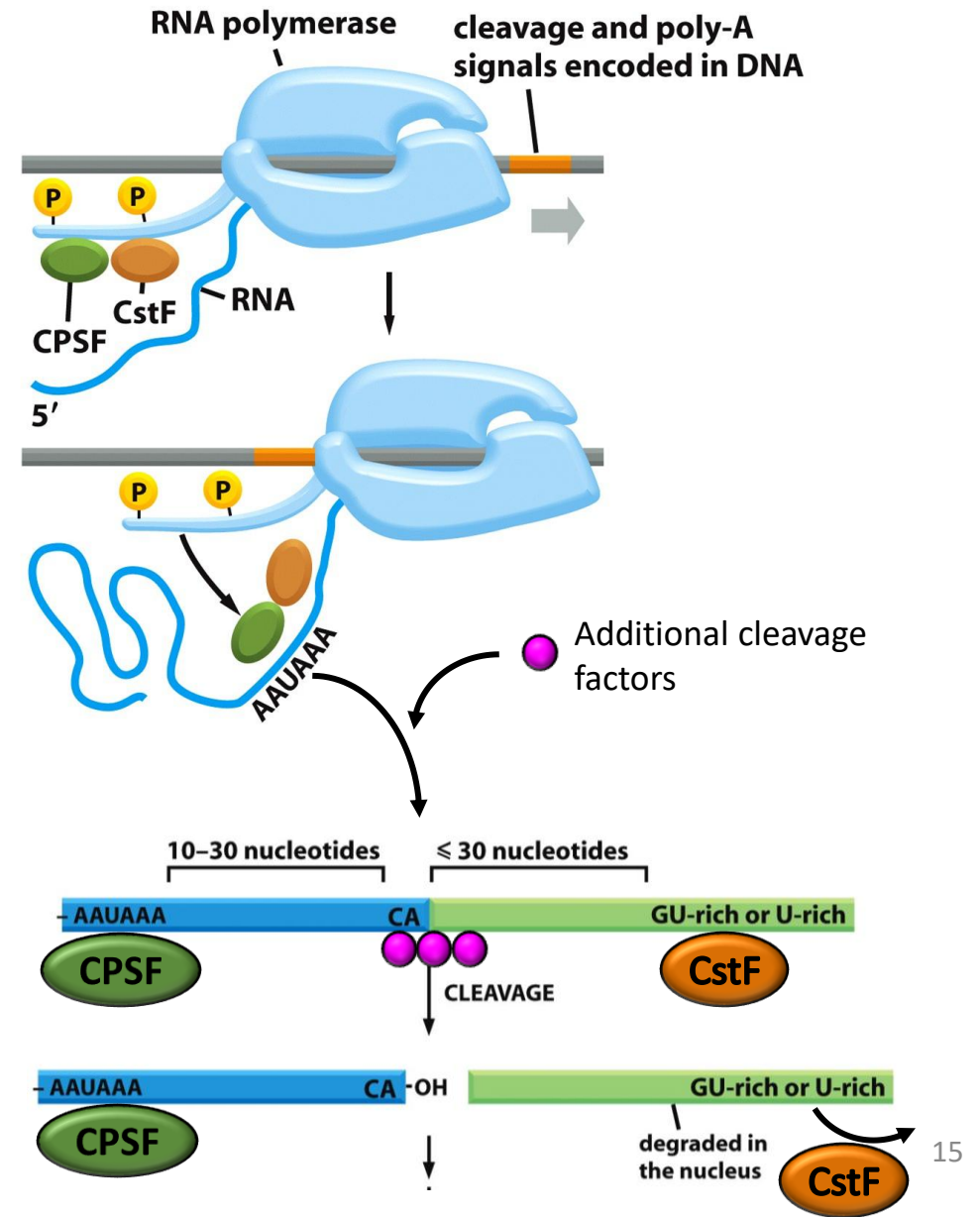
The CBC & eIF-4E/eIF-4G block the access of de-capping enzymes to the cap. This increases the $t_{1/2}$ of the mRNA (the life time of the transcript in the cytosol)



3' mRNA Processing in Eukaryotes- Cleavage

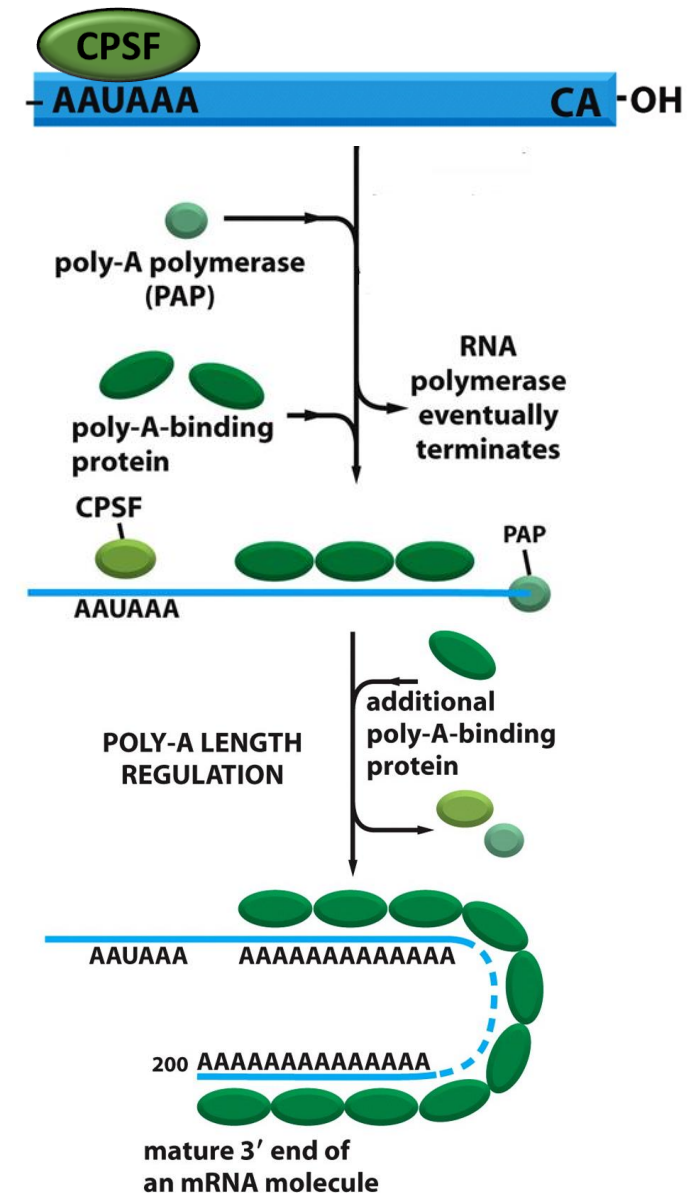
The modification of the 3' end of the RNA is accomplished by several enzymes associated with RNA polymerase II that bind to specific sequences on the RNA

- 1) Cleavage and polyadenylation specificity factor (**CPSF**) binds to the hexamer AAUAAA (the polyadenylation signal).
- 2) Cleavage stimulating factor F (**CstF**) binds the GU-rich element beyond the cleavage site.
- 3) Cleavage factors ● bind to the CA sequence at the cleavage site.



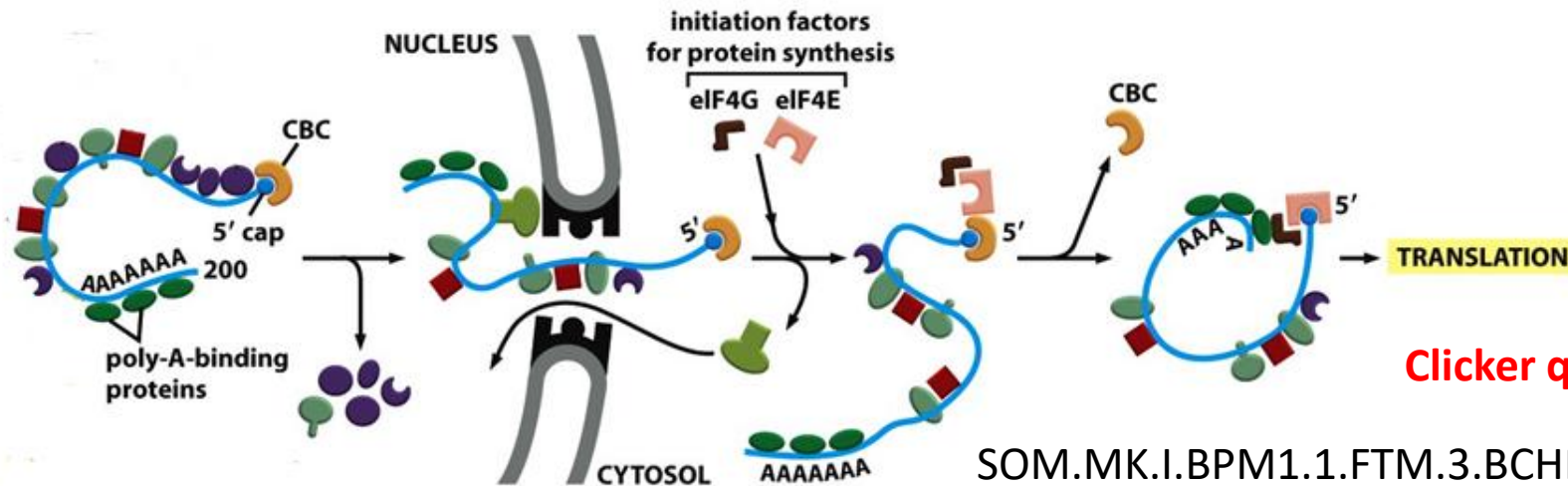
3' mRNA Processing in Eukaryotes- Polyadenylation

1. Poly-A-polymerase (PAP) adds approximately 200 A nucleotides to the 3' end produced by the cleavage
2. Poly-A Binding Proteins (PABP) binds to the poly-A tail and assist in directing translation by the ribosome
3. The cleaved fragment of the RNA is degraded in the nucleus.



Functions of 3' mRNA Polyadenylation

- Increases the $t_{1/2}$ of the mRNA by protecting it from enzymatic degradation in the cytoplasm
- Poly(A)-binding protein (PABP) binds to poly(A) tracts protecting mRNAs from ribonuclease attack
- PABP interacts with e-IF 4G, and this interaction is thought to lead to the stimulation of translation of eukaryotic mRNAs
- Aids in transcription termination i.e., facilitates transcription termination.
- Aids in export of the mRNA from the nucleus



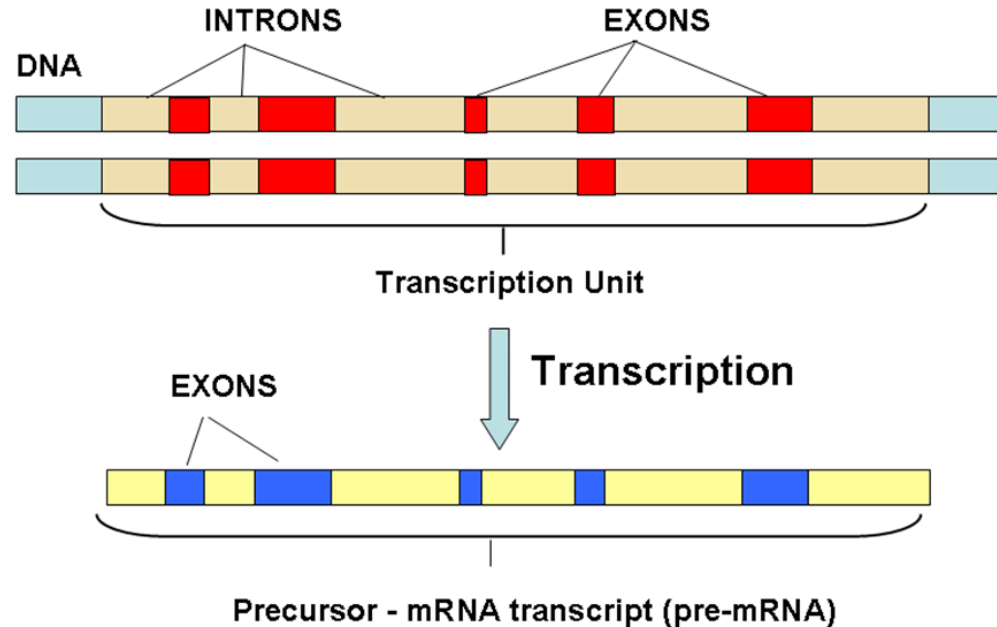
Clicker question next

Removal of Introns by RNA splicing

- Protein coding sequences (exons) of eukaryotic genes are typically interrupted by non-coding sequences (introns)
- Human introns range from about 10 to over 100,000 nucleotides

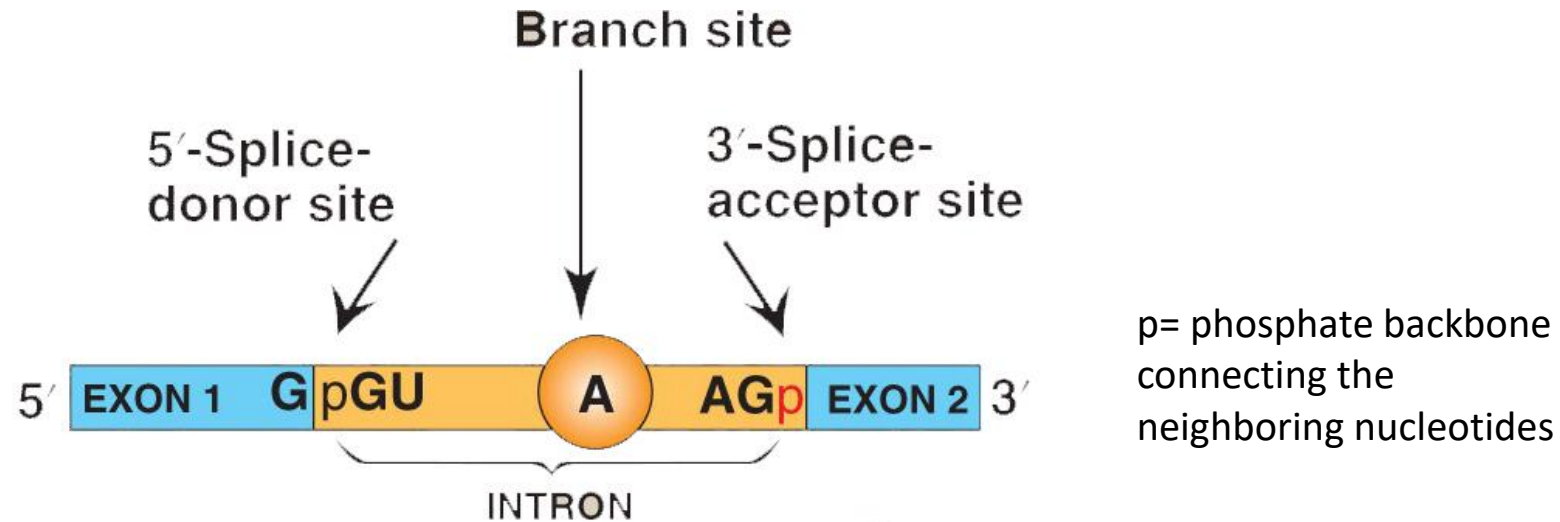


Expressed Sequence = exons
Intervening Sequence = introns



- Both introns and exons are transcribed to mRNA
- Intronic sequences are then removed by **splicing**

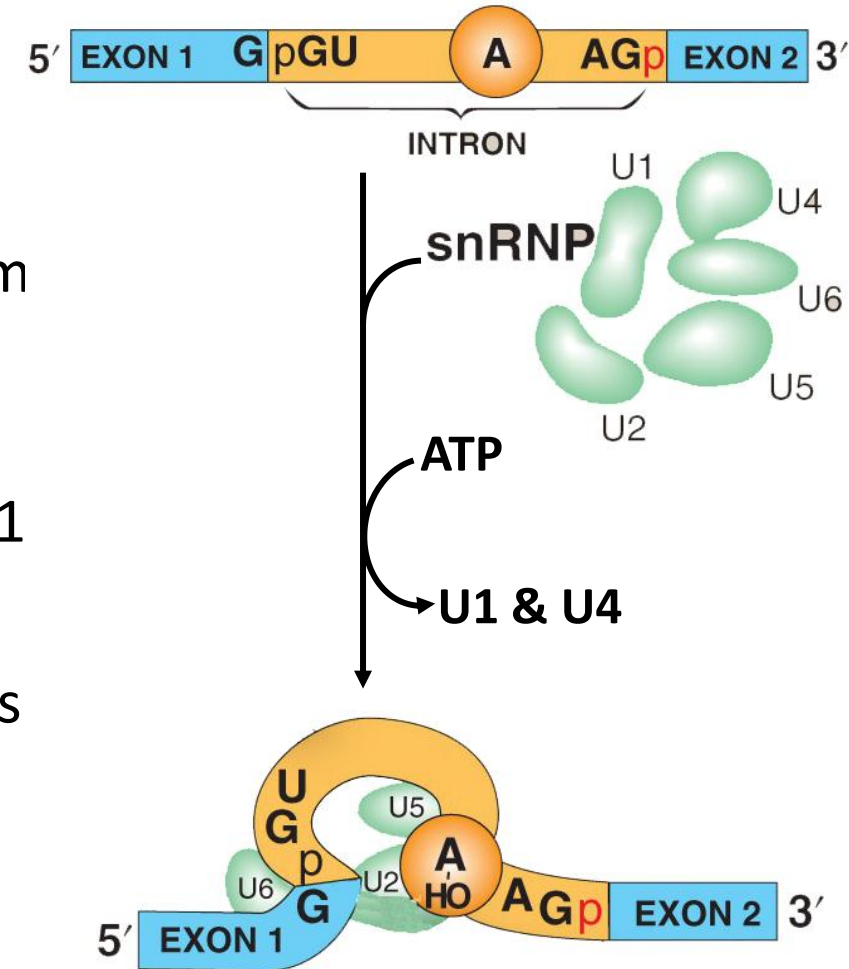
Most mRNA splicing takes place during transcription



- Splicing takes place at splice junctions located at the 5' and 3' end of the intron
- The junctions have conserved consensus sequences (6-8 nt)
- The 5' end of the intron consensus sequence is always **GU**
- The 3' end of the intron consensus sequence is always **AG**
- The branch site is always '**A**' which is found 18-38 nt upstream from the 3' end of the intron
- Each splicing event removes 1 intron

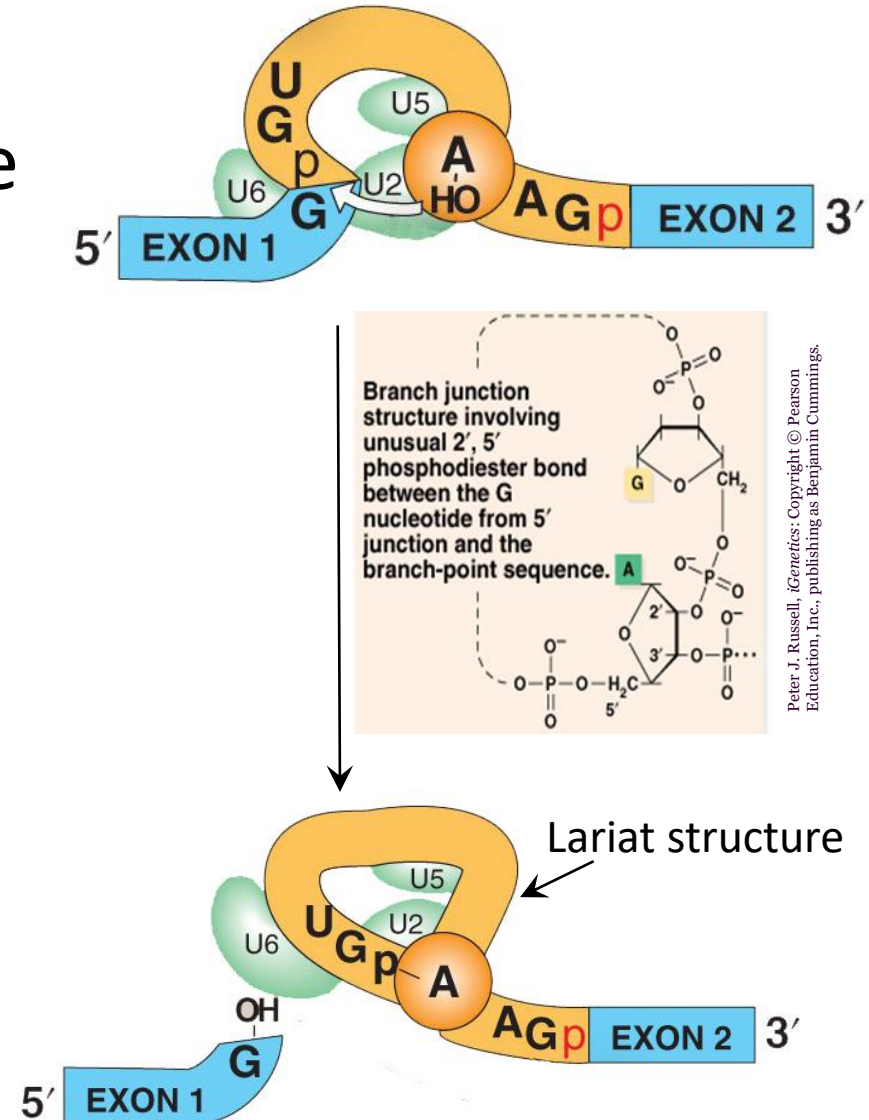
Details of Intron Removal from a pre-mRNA molecule

- 5 small nuclear ribonucleoprotein particles (snRNP) and numerous other splicing factors mediate splicing
- The snRNPs are U1, U2, U4, U5 and U6 which form base pairs with consensus sequences at each end of the intron (intron-exon junctions)
- After the 5 snRNPs bind the primary transcript, U1 and U4 leave, thus activating the spliceosome
- U2, U5 and U6 interact to bring neighboring exons into correct alignment allowing for **two trans-esterification reactions** to occur



Details of Intron Removal from a pre-mRNA molecule

- The 2'-OH of the branch site A attacks the 5' phosphate (p) at the splice donor site (5' end of the intron) forming an 2'→5'-unusual phosphodiester bond
- This reaction produces a lariat structure
- This reaction also leaves a free 3' hydroxyl at the end of exon 1 which will participate in the next reaction



Details of Intron Removal from a pre-mRNA molecule

- The newly freed 3'-OH exon 1 then attacks the 5'-phosphate (**p**) at the splice-acceptor site to form a phosphodiester bond that joins exons 1 and 2
- The excised intron is released as a lariat which is typically degraded
- The mature messenger RNA molecules pass into the cytosol through pores in the nuclear membrane

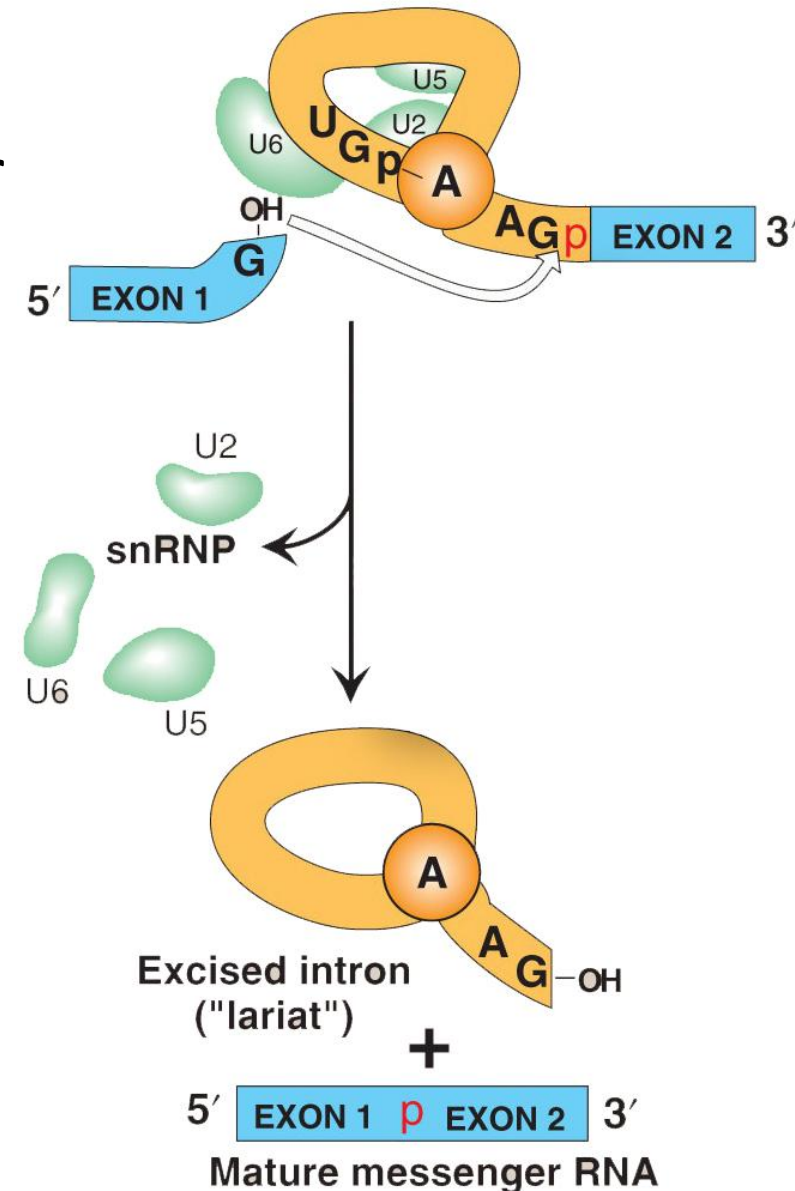
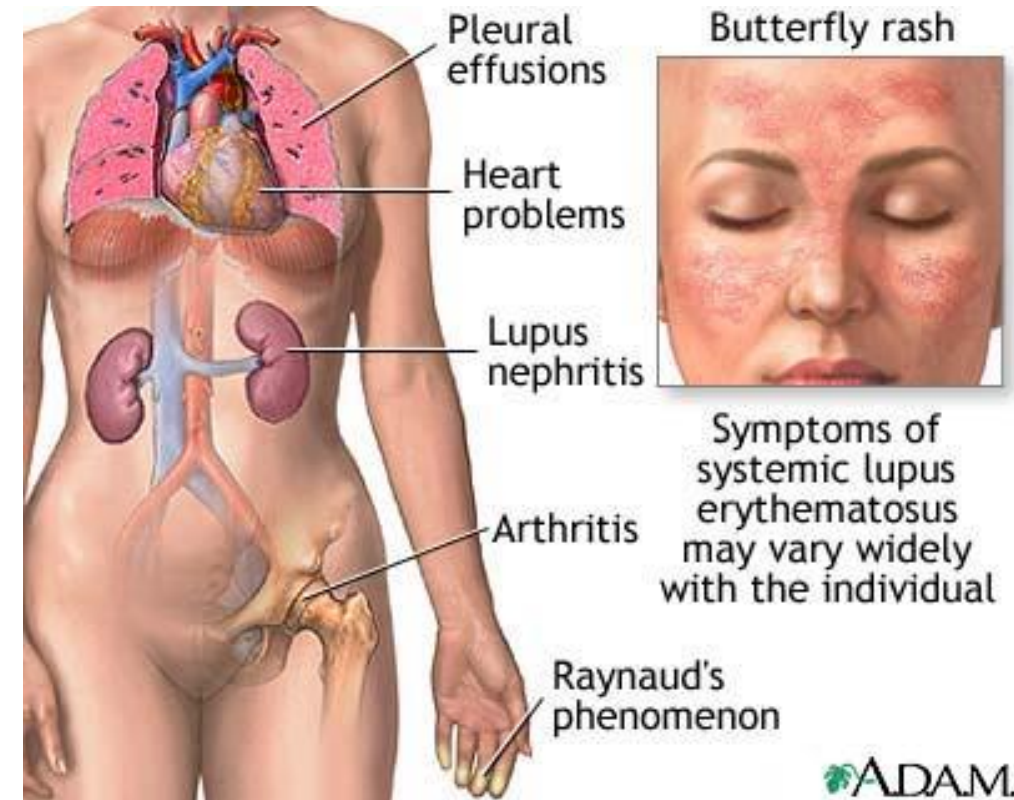


Figure 31.18 Lippincott Biochemistry 7th ed.

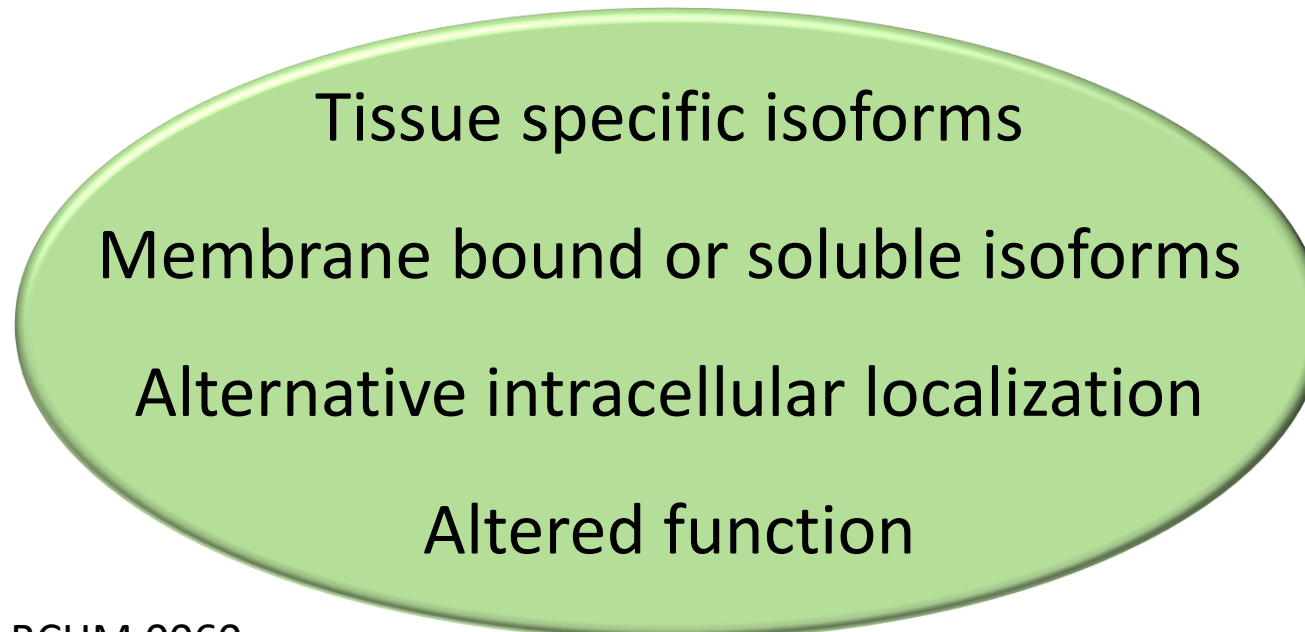
Systemic Lupus Erythematosus

- Conditions that negatively affect the components of the spliceosome can have clinical effects
- Lupus is an autoimmune disease with multiple symptoms including extreme fatigue, arthritis, fever skin rashes and kidney problems
- Individuals with this condition possess antibodies that react against their own nuclear proteins such as the U1 RNA component of the spliceosome, histones and topoisomerases



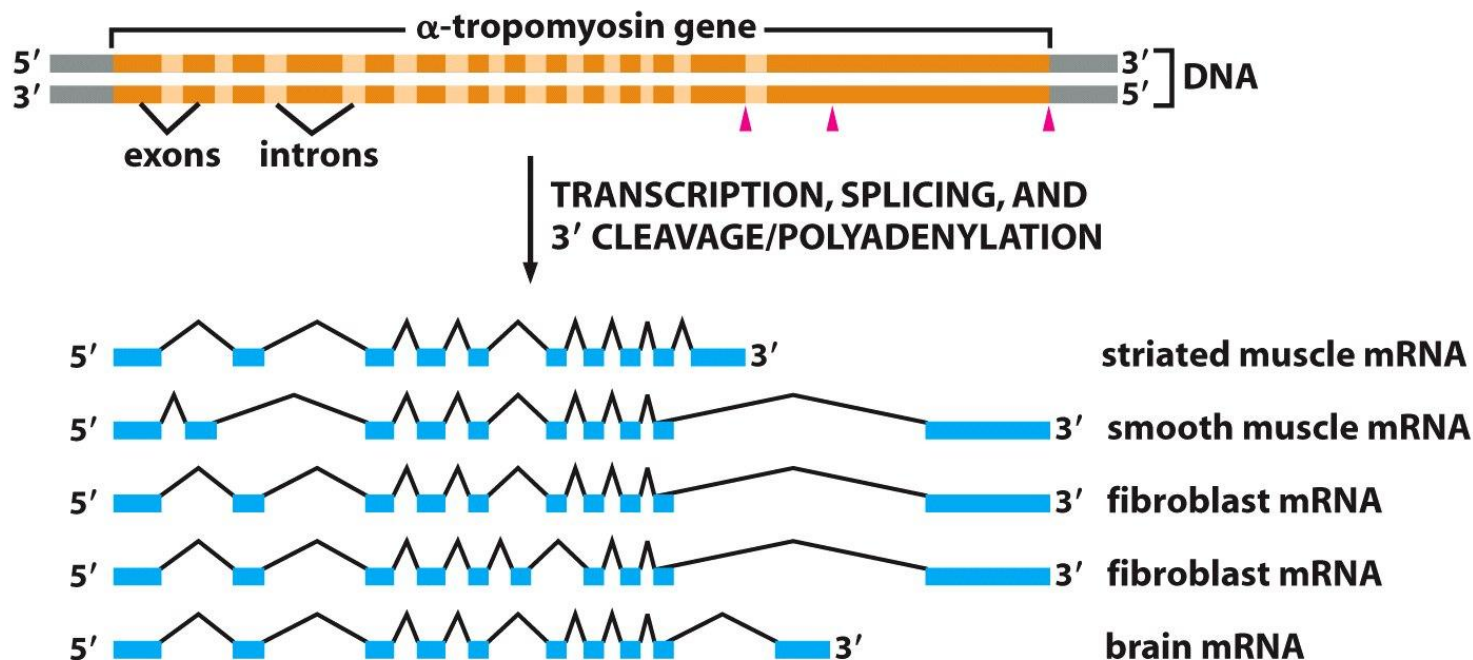
Increasing the molecular diversity of mRNAs & proteins

- Biological diversity can be increased by carrying out:
 - Alternative splicing and/or
 - RNA editing
- Allows a number of different proteins to be produced from the same gene



Alternative Splicing

- RNA can be spliced in different ways to produce different species of mRNA in a process called *alternative splicing*
- Alternative RNA splicing can produce different forms of a protein from the same gene
- ~60% of human genes are spliced in alternative ways producing a different set of mRNAs



α -Tropomyosin is a protein with many isoforms: from regulating actin filaments in muscle cells (muscle contraction) and non-muscle cells (cell division and shape)

Figure 6-27 Molecular Biology of the Cell 5/e (© Garland Science 2008)

Control of Alternative RNA Splicing

- 1) Alternative splicing can occur due to intron sequence ambiguity. This may be due to 'weak' splice site sequences and therefore splicing choice occurs by chance
- 2) Alternative splicing can also be directed by negative and positive control via proteins that bind to splice sequences and either repress or activate splicing at that site

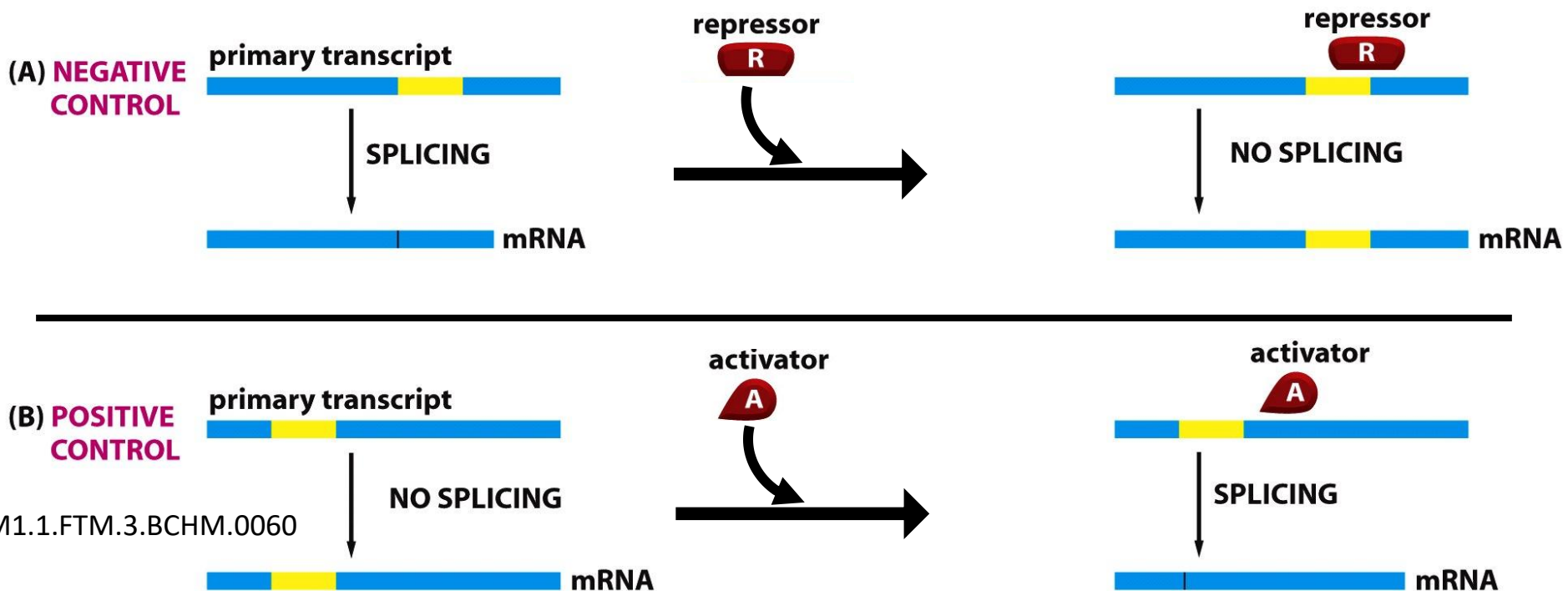


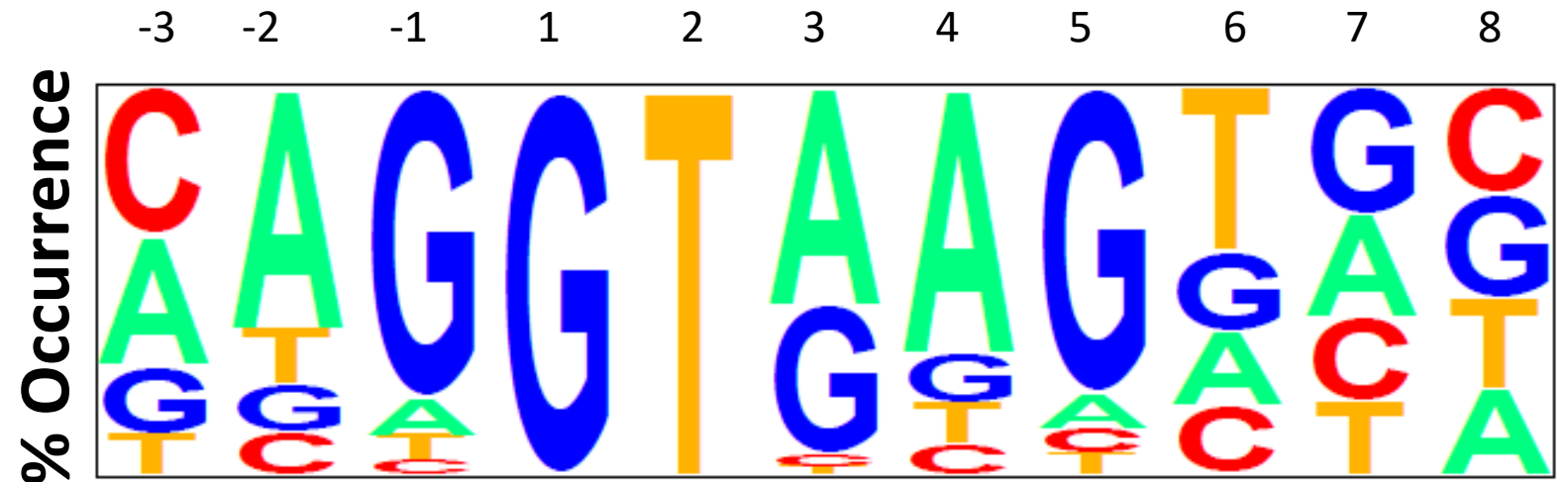
Figure 7-96 Molecular Biology of the Cell 5/e (© Garland Science 2008)

Mutations in genes that affect the sequence of splice donor/acceptor sites or that create new splice donor/acceptor sites can cause clinical effects

Conserved Regions for RNA Splicing

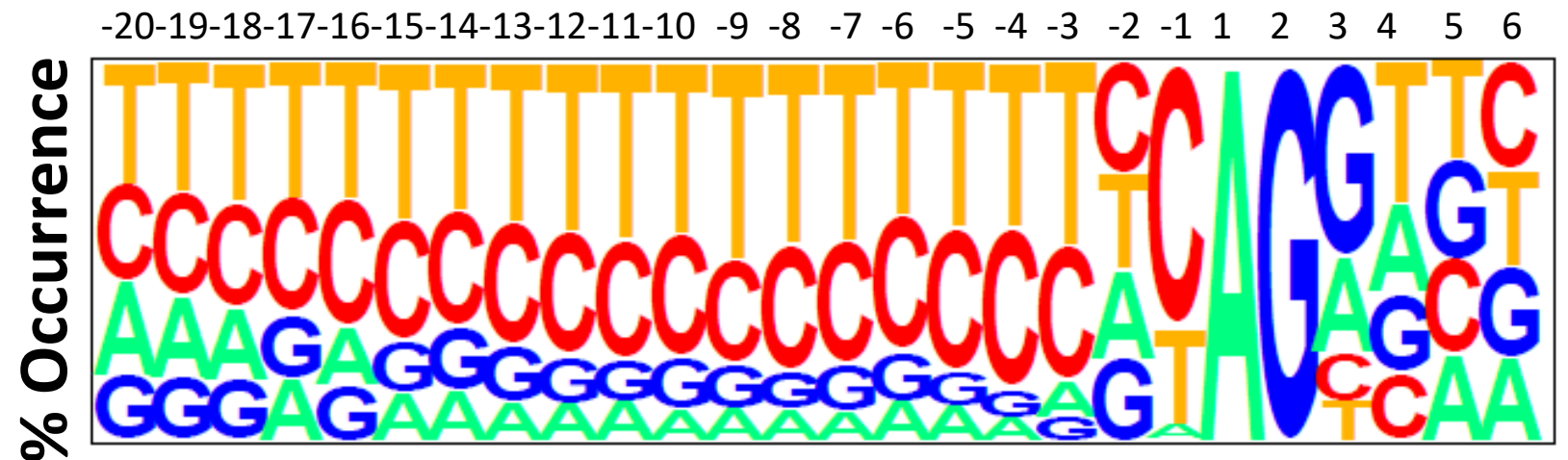
Donor Site:

This figure shows the probability of the conserved sequence of the donor site. Notice the absolutely conserved GT.



Acceptor Site:

This figure shows the probability of the conserved sequence of the acceptor site. Notice the absolutely conserved AG.



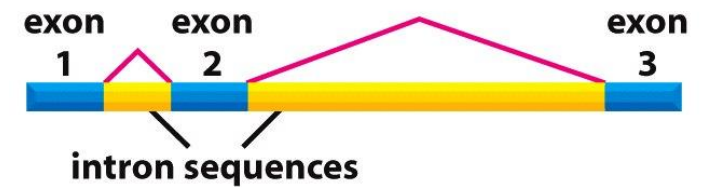
β -thalassemia

- Many individuals with β -thalassemia have mutations in the β -globin gene that generate additional splice sites within the mRNA, two examples shown
- **Example B** shows a mutation in a normal acceptor site in intron 1 so that exon 2 is skipped
- **Example C** shows a mutation in the normal acceptor site of intron 2 but there is a sequence upstream which is a weak acceptor site which is now used:

Activation of a 'cryptic splice site'

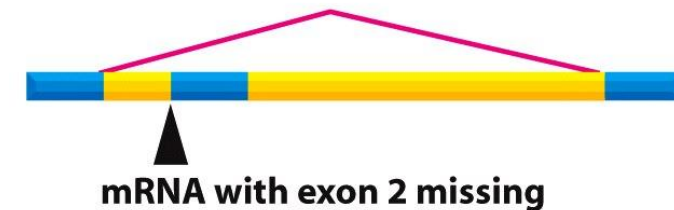
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(A) NORMAL ADULT β -GLOBIN PRIMARY RNA TRANSCRIPT



normal mRNA is formed from three exons

(B) SOME SINGLE-NUCLEOTIDE CHANGES THAT DESTROY A NORMAL SPLICE SITE CAUSE EXON SKIPPING

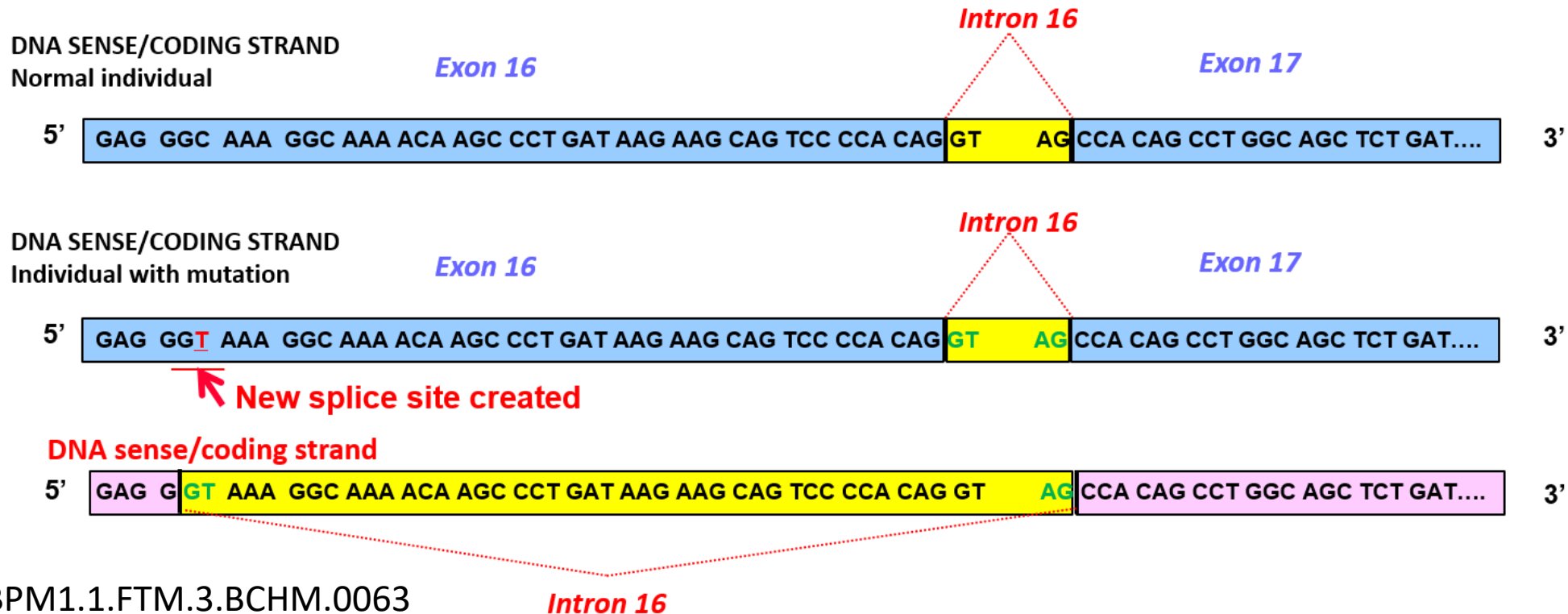


(C) SOME SINGLE-NUCLEOTIDE CHANGES THAT DESTROY NORMAL SPLICE SITES ACTIVATE CRYPTIC SPLICE SITES



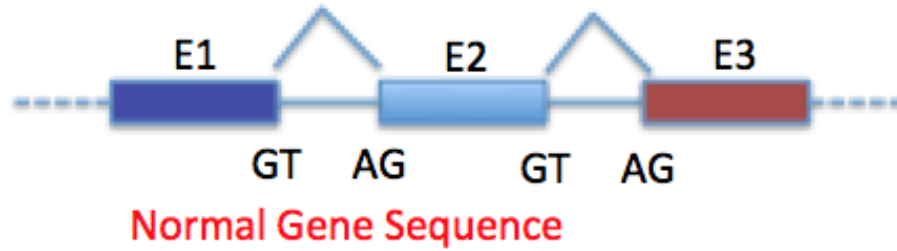
Limb Girdle Muscular Dystrophy

- Some patients with Limb Girdle Muscular Dystrophy possess a mutation in the calpain-3 gene that generates a new splice site within exon 16
- This results in a shorter calpain-3 mRNA that is missing many base pairs in exon 16
- Mutation interrupts a codon causing downstream codons to be out of frame and thus producing a defective protein

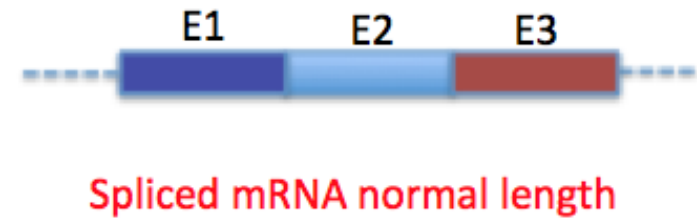


The effect of mutation on pre-mRNA splicing

DNA sequence showing pre-mRNA splicing

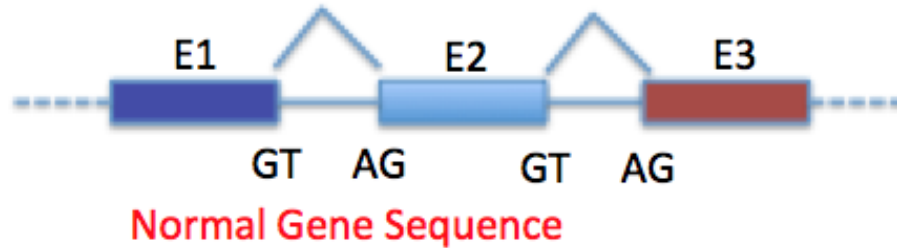


Mature mRNA after splicing

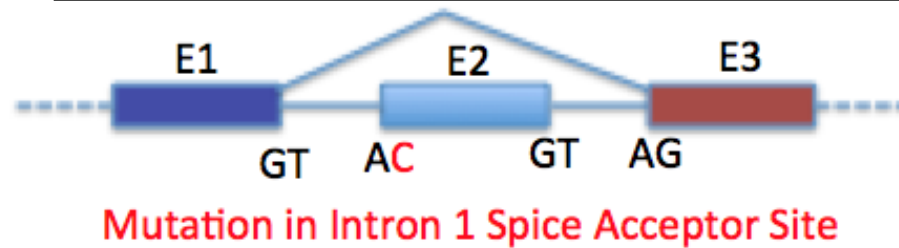
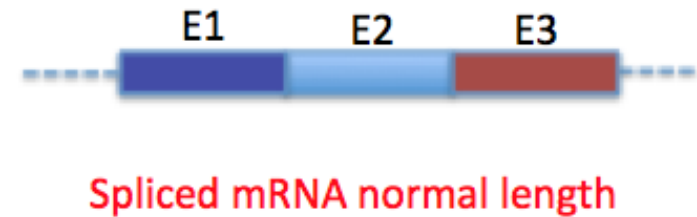


The effect of mutation on pre-mRNA splicing

DNA sequence showing pre-mRNA splicing

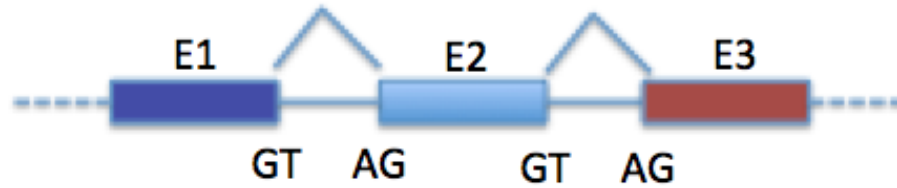


Mature mRNA after splicing



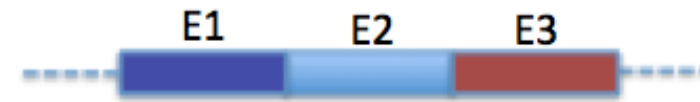
The effect of mutation on pre-mRNA splicing

DNA sequence showing pre-mRNA splicing

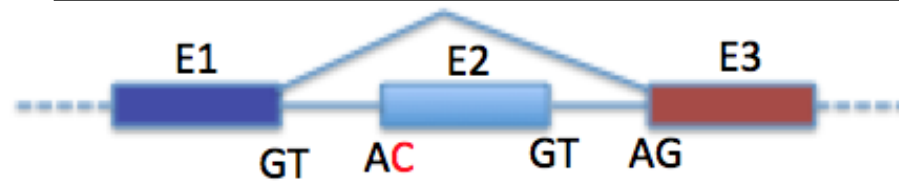


Normal Gene Sequence

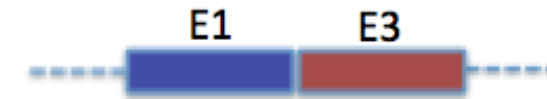
Mature mRNA after splicing



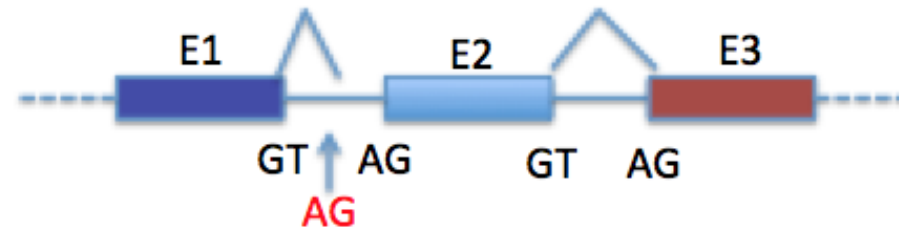
Spliced mRNA normal length



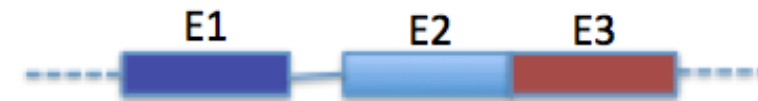
Mutation in Intron 1 Splice Acceptor Site



Spliced mRNA shorter length



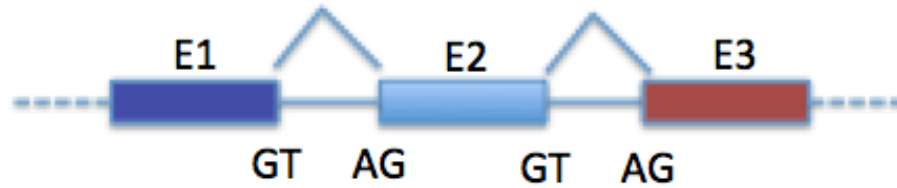
Mutation in Intron 1 introduces a splice acceptor site



Spliced mRNA longer length

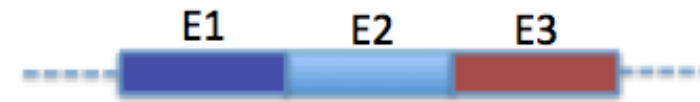
The effect of mutation on pre-mRNA splicing

DNA sequence showing pre-mRNA splicing

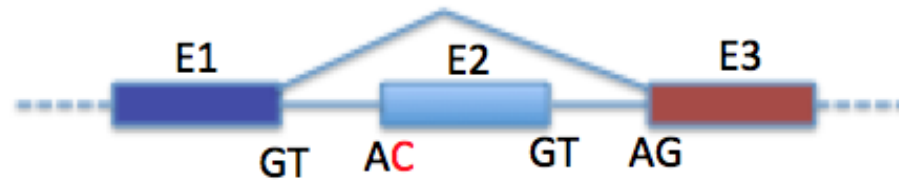


Normal Gene Sequence

Mature mRNA after splicing



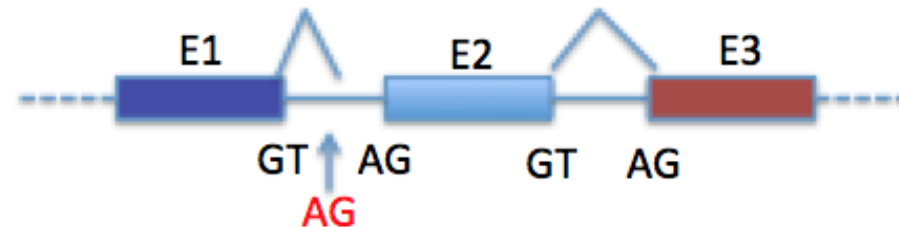
Spliced mRNA normal length



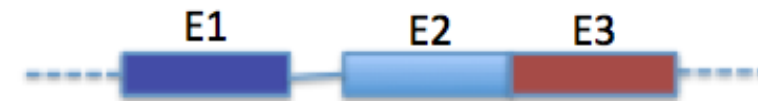
Mutation in Intron 1 Splice Acceptor Site



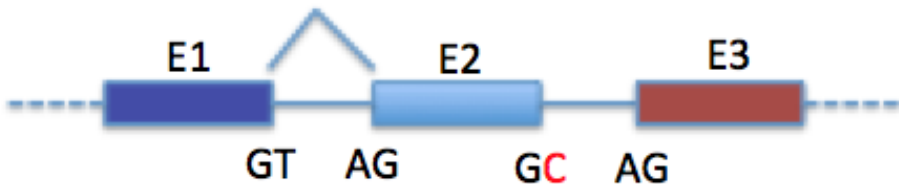
Spliced mRNA shorter length



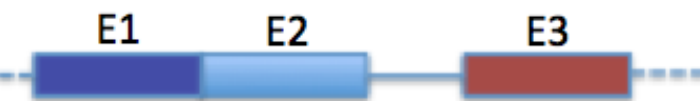
Mutation in Intron 1 introduces a splice acceptor site



Spliced mRNA longer length



Mutation in Intron 2 Splice Donor Site



Spliced mRNA longer length

Both tRNA and rRNAs are extensively chemically modified to enable formation of specific structural features and possible catalytic activities

Chemical modification of mRNA is known as RNA Editing
Used by humans and many other species

RNA editing by chemical modification of mRNA

Chemical modification of mRNA



Coding properties of transcript are altered



Results in alteration in the amino acid sequence of the specified protein & generation of functionally different proteins

There are two types of editing

- Adenosine to Inosine (A-I)
- Cytidine to Uridine (C-U)

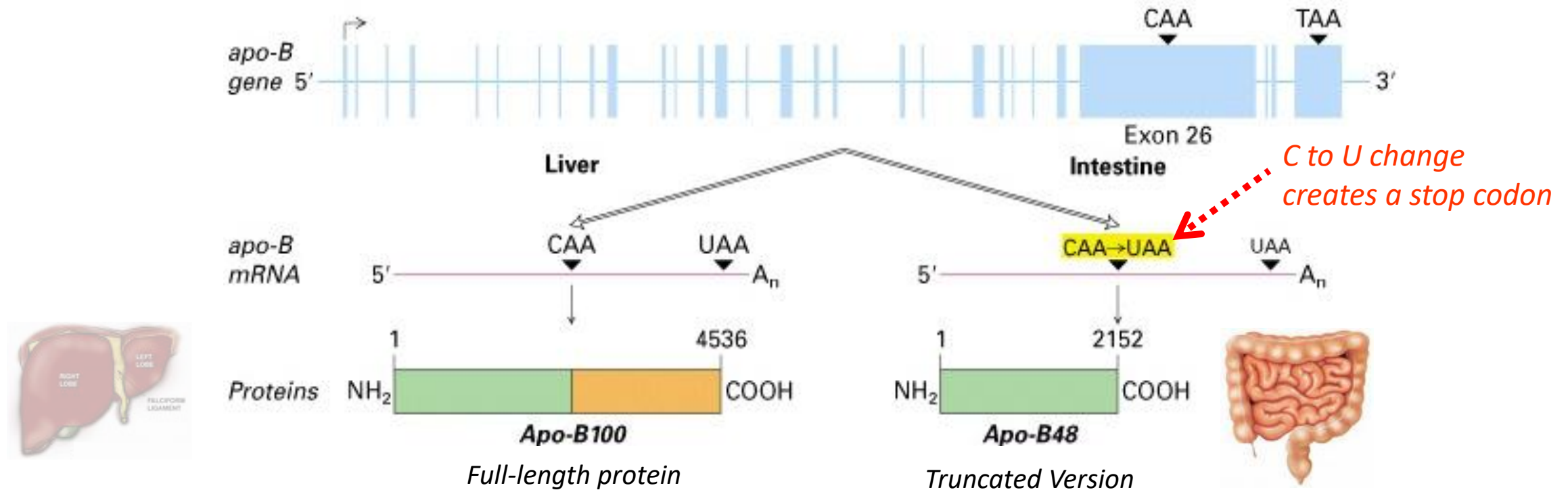
RNA editing by chemical modification of mRNA

RNA editing is a process in which information changes at the level of mRNA

It is revealed by situations in which the coding sequence in an RNA differs from the sequence of DNA from which it was transcribed

In mammalian cells there are cases in which a substitution occurs in an individual base in mRNA, causing a change in the sequence of the protein that is coded

C-U editing in Apolipoprotein-B mRNA by Cytidine Deaminase

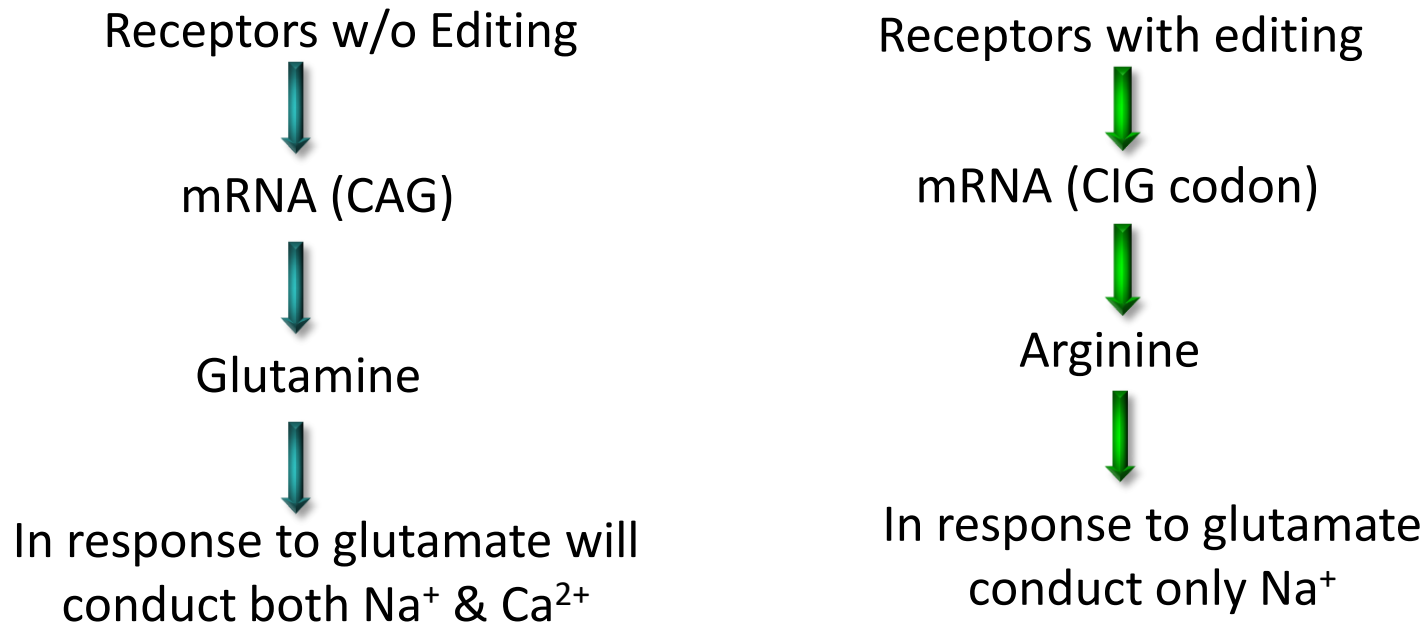


In mammals, the *apo-B* gene encodes two alternative forms of the apolipoprotein B: *Apo-B100* (liver) and *Apo-B48* (intestine)

In the intestine, the *apo-B* mRNA is edited so that a premature stop codon is produced (CAA→UAA) leading to the synthesis of the shorter Apo-B48

This editing is accomplished by a ***cytidine deaminase enzyme***

A-I editing: Glutamate receptor mRNA in Brain



Editing of the glutamate receptor involves a single base change from A to I (inosine). This editing results in a codon change within the mRNA from glutamine to arginine

The editing affects an amino acid located in the inner wall of the ion channel and alters Ca²⁺ permeability.

This editing is a requirement for proper brain development.

This editing is accomplished by the enzyme ***adenosine deaminase***.

mRNA Processing Summary

- Prokaryotes don't do it!
- Eukaryotes add a 7-methyl-guanosine cap to the 5' end of the mRNA & a poly A tail to the 3' end to form pre-mRNA.
- Pre-mRNA is spliced to remove introns with the aid of the spliceosome – forms mature mRNA.
- Alternative splicing occurs to increase diversity of mRNA & expressed protein.
- RNA editing can also be used to diversify mRNA, although this is less common.
- Abnormal splicing can generate abnormal proteins which can cause disease e.g. β -thalassemia

Thank you for your attention

The End

